

JY Liu

Solutions Architect | Technical Lead | Senior Full-Stack Engineer

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Summary

- Solutions Architect and Technical Lead with 12+ years designing and delivering enterprise, fintech, SaaS, and integration-heavy platforms across Asia and the US.
- Combines hands-on full-stack engineering with architecture ownership across cloud infrastructure, APIs, distributed systems, payments, and DevOps — proven delivery across regulated and mission-critical environments, including global payment infrastructure supporting 235,000+ users and consular platforms for global diplomatic clients.
- Recent focus on AI-enabled product engineering and agentic development workflows, complemented by ongoing MBA studies in Business Analytics at the University of Malaya.

Experience

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| Vailley AI Solution Pte. Ltd. , Senior Web Developer (Technical Lead) | Remote (Singapore)
Jun 2025 – Present |
| <ul style="list-style-type: none">• Led solution architecture and technical delivery across VFS Global's visa and consular platforms — translating business and security requirements into cloud-native application, integration, and data architectures while coordinating frontend, backend, and platform delivery and remaining hands-on across the stack.• Delivered high-security Consular Dashboard with granular RBAC and ABAC access controls, an ADR portal with structured document submission, ETL-backed operational visibility, a digital corporate visa platform for high-volume agent workflows, and SAP Concur to OneVasco integration — supporting multi-country diplomatic operations for VFS Global. | |
| Shift Markets , Senior Software Engineer | Remote (NY, US)
Jan 2023 – Present |
| <ul style="list-style-type: none">• Lead engineer for CXM Direct (worldwide multi-asset trading platform) and CRM initiatives, owning backend delivery with ThinkPHP and frontend execution in TypeScript.• Engineered a Unified Payment Gateway integrating 12 PSPs across 12+ regions and 17+ currencies, supporting 235,000+ registered users across CXMD (Singapore) and CXMT (Tokyo) trading platforms.• Strengthened system reliability and delivery velocity through architectural cleanup, perf tuning, and production-oriented engineering practices. | |
| InternetBrands , Senior Web Engineer | Remote (California, US)
Apr 2023 – Sept 2024 |
| <ul style="list-style-type: none">• Maintained and improved the Dentrix Hub Forms (Online Form for Dentrix & HOP) Laravel backend — patient form management platform for dental practices, focused on reliability, performance, and scale.• Introduced PHP linting and unit testing standards, growing test coverage from 0% to 90% across the codebase.• Managed Kubernetes deployment with Helm and Rancher for the backend. | |
| MaiDaLv , Senior Full Stack Developer | Remote (Guangzhou, China)
Feb 2025 – Apr 2025 |
| <ul style="list-style-type: none">• Delivered Spring Boot, UniApp, Vue.js, and Python-based product workflows covering subscription management, WeChat payment integration, and IP-risk scanning. | |
| Cake Experiential Communications Asia , Solutions Architect | Remote (KL, Malaysia)
Apr 2021 – May 2023 |
| <ul style="list-style-type: none">• Led end-to-end architecture and delivery for Vue.js and Laravel platforms used in regional enterprise and campaign engagements.• Introduced engineering standards — TDD, API governance, and backend scaling practices — for client-critical systems. | |
| Glints , Web Software Engineer | Remote (Singapore)
Jan 2022 – Dec 2022 |
| <ul style="list-style-type: none">• Built full-stack recruiter tooling for Superpowered using Next.js and Koa.js, with emphasis on workflow efficiency and usable internal product design.• Strengthened dashboard responsiveness and release quality through improved testing and deployment automation. | |

InnoLab Global , Senior Information Technology Consultant		Remote (KL, Malaysia) Sept 2018 – Present
- Delivered cross-platform products across mobile (Ionic, Flutter, React Native), web (Angular, React, Next.js, Vue.js), and backend (Laravel, NestJS, Spring Boot, Node.js, Java, Python) stacks on cloud infrastructure (AWS, Firebase, GCP) across a diverse client portfolio, while automating cloud delivery workflows and mentoring junior developers. - Shipped the Forest City Super App on a Turborepo + Next.js + Prisma + PostgreSQL + Auth.js stack covering community, events, payments, partner portal, and admin console features for residents and partners. - Architected and led delivery for AvaEra's compensation platform — modernized legacy stack into pnpm + Turborepo monorepo with Next.js, Prisma, PostgreSQL, and a 15-service reward engine covering direct sponsor, team bonus, pool sharing, profit distribution, and merchant commissions. - Built and shipped HappyAI for Gembira Bina Sdn Bhd, an AI-powered tender document extraction and validation platform — Next.js + FastAPI + SQLite with Groq LLM primary / Ollama fallback, confidence-badged review UI, and deterministic batch ingest pipeline.		
Supahands , Full Stack Developer		Remote (KL, Malaysia) Feb 2018 – Jun 2018
Built full-stack features for Supahands' distributed-team platform supporting crowdsourced microtask workflows.		
The Consociation LLP , Information Technology Consultant		Remote (Singapore) Sept 2016 – Apr 2020
Led full-stack delivery for products including iHeartCity and Swiiipe using Angular, NestJS, GraphQL, and Firebase with AWS-based DevOps automation.		
Buddy Holdings Pte Ltd , Senior Information Technology Consultant		Remote (Singapore) Sept 2016 – Jan 2024
- Architected and led delivery of the Buddy System Asia MLM platform serving 2,000+ active members across Singapore, Malaysia, China, and Indonesia — Angular + Laravel + MySQL + Kafka covering hierarchy management, point calculations, point redemption, inventory tracking, and family tree updates at scale. - Built agent-facing UI for real-time tracking of progress, points, and family trees across the multi-country user base.		

Earlier Experience

Roles

Nexstream Sdn Bhd (Hybrid App Developer, 2014–2015, KL) · Netccentric Pte Ltd (Software Developer, 2015, Singapore) · Fuji Xerox Pte Ltd (Embedded Software Engineer, 2016, Singapore)

Education

Master of Business Administration (Expected Oct 2026)	University of Malaya , Business Analytics	Kuala Lumpur, Malaysia Oct 2023 – Oct 2026
Master of Computer Science	University of Malaya , Mobile Cloud Computing	Kuala Lumpur, Malaysia Jan 2011 – Jan 2013
Bachelor of Computer Science	Universiti Sains Malaysia , Software Engineering	Malaysia Jan 2008 – Jan 2011

Skills

Full Stack Engineering: TypeScript, JavaScript, PHP, Python, Java, Laravel, ThinkPHP, React, Next.js, Angular, Vue.js, NestJS, Node.js, Spring Boot, Flutter, Ionic

Architecture, Cloud & DevOps: Solution architecture, AWS, GCP, Firebase, Docker, Kubernetes, Helm, Rancher, CI/CD, deployment automation, integration-heavy platform delivery

Backend, Data & Integration: PostgreSQL, MySQL, Redis, Apache Kafka, GraphQL, REST APIs, microservices, ETL workflows, event-driven and serverless systems

AI Engineering & Agentic Systems: LLM orchestration, RAG pipelines, structured extraction, prompt engineering, custom GPTs, Groq, Ollama

Languages

English, Malay, Cantonese: Professional

Mandarin: Native